

# Financial Technology

HW3 Deadline: 2025/11/3 23:59

## Problem :

1. Collect the TAIEX from 2019/09/30 to 2025/10/03 (Day Bar). The data should include open, close, high, low, volume.
2. Apply the triple-barrier method to label the collected data. The upper bound is set as 4% and labeled as 1 once it is touched. The lower bound is set as 2% and labeled as 2 once it is touched. The vertical barrier is set as 20 days and labeled as 0 once it is touched.
3. Use “close price” to calculate 8 technical indicators of collected data. The technical indicators are:
  - A. Bias ratio of moving average: 5-days, 10-days, 20-days, 60-day.
  - B. RSI: 14
  - C. MACD(快線 DIF), MACD signal(慢線), MACD histogram(柱狀)
  - D. Save problem 1, 2, 3 to a csv.
4. Use the collected data in problem 1 and all technical indicators in problem 3 to calculate the 2 PCA component.
  - A. Plot the explained variance ratio and its cumulative sum.
  - B. Plot data by class labeled in problem 2.
  - C. Discuss pros and cons of PCA method in this dataset.

## 技術指標說明：

- **Bias ratio of moving average:**  
 $((\text{價格} - \text{MA}) / \text{MA})$
- **RSI:**  
EMA 改為 SMA
- **MACD:**  
參數預設:(12, 26, 9)

## Grading Policy:

1. In the programming assignment, the **code(.ipynb or .py)**, **report(.pdf)**, **hw3.csv**, and **Readme** file should all be compressed into a **ZIP** file named 學號\_姓名 and upload to eeclass website. If your code contains external numerical or image data etc. instead of data stream, it should be contained in the ZIP too. The Readme file should explain how to run your code.
2. The programming language that can be used on this assignment is Python. Built-in libraries or functions are allowed to be used. Please integrate all your code into a main file.
3. According to the question requirement, the result should be contained in your **report**. **Completeness of responses will affect your score (Explanation, Process, Results, and Discussion).**
4. 30% off for late submission within one week, not accepted after one week.
5. Discussions are encouraged, **but plagiarism is strictly prohibited.**

